

MASS-ENG-023

Communications Engineering Specification

Version 1.0

Engineering Objective

Communications implements the constitutional capability responsible for enterprise communication stewardship. It governs communication strategy, channel orchestration, message delivery policies, and communication intelligence while preserving constitutional ownership boundaries.

Engineering Preface

Communications is not the owner of enterprise information. Other capabilities produce enterprise intelligence; Communications ensures that intelligence reaches the right audience, through the right channel, at the right time, using the appropriate enterprise standards. Communications stewards delivery—not business intent.

Purpose & Scope

Communications governs enterprise messaging, channel selection, communication policies, templates, delivery orchestration, acknowledgements, communication history, and effectiveness analysis. Out of Scope: relationship stewardship, customer experience ownership, growth strategy, operational execution, dispatch coordination, and financial processing.

Architecture

Core Components: Communication Service, Channel Service, Template Registry, Preference Service, Delivery Orchestration Service, Communication History Repository, Policy Service, Analytics Service, Subscription Service, Communication Intelligence Service.

Engineering Requirements

Functional: orchestrate communications, honor channel preferences, manage templates, track delivery, evaluate effectiveness, support acknowledgements, publish communication intelligence.

Non-Functional: secure, observable, resilient, accessible, privacy-aware, constitutionally compliant.

Dependencies & Responsibilities

Requires Enterprise Core, Identity, Security. Uses Workflow, Documents, AI Orchestration, Notifications, Observability, Persistence, Error, Configuration, API, Integration, Relationship Command, Customer Experience, Growth, Operations, Dispatch, Finance. Provides Communication Intelligence, Channel Governance, Delivery Strategy, Communication Analytics. Owns communication stewardship; does not own relationship intent, customer experience, growth strategy, operational execution, dispatch, or finance.

Interfaces, Example Flow & Definition of Done

Interfaces: `ComposeCommunication()`, `SelectChannel()`, `DeliverCommunication()`, `RecordAcknowledgement()`, `EvaluateEffectiveness()`, `PublishCommunicationAnalytics()`. Example Flow: Relationship Command identifies audience -> Customer Experience defines intent -> Growth requests campaign -> Communications selects channels and orchestrates delivery -> Notification Engine delivers -> Analytics evaluates effectiveness. Definition of Done: Communications consistently governs enterprise messaging and channel orchestration while preserving constitutional ownership boundaries and producing measurable communication intelligence.